JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY, NOIDA



Full Stack Development Lab

(24B15CS314)

Topic: Exam Management System

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1. Problem Statement & Objectives

A. Problem Statement

In educational institutions, the process of creating exam seating arrangements and generating individual admit cards is often carried out manually. This approach is time-consuming, prone to errors, and difficult to manage when the number of students is large. Students also face inconvenience in quickly finding their allotted seat and room details before the examination.

B. Objectives

- To develop a web-based Exam Management System using Python and Django.
- To automatically generate seating arrangements based on student data and available rooms.
- To generate individual admit cards in PDF format.
- To allow students to search their seat details in real-time using AJAX.
- To provide basic reports (room-wise student list) in CSV and PDF format.

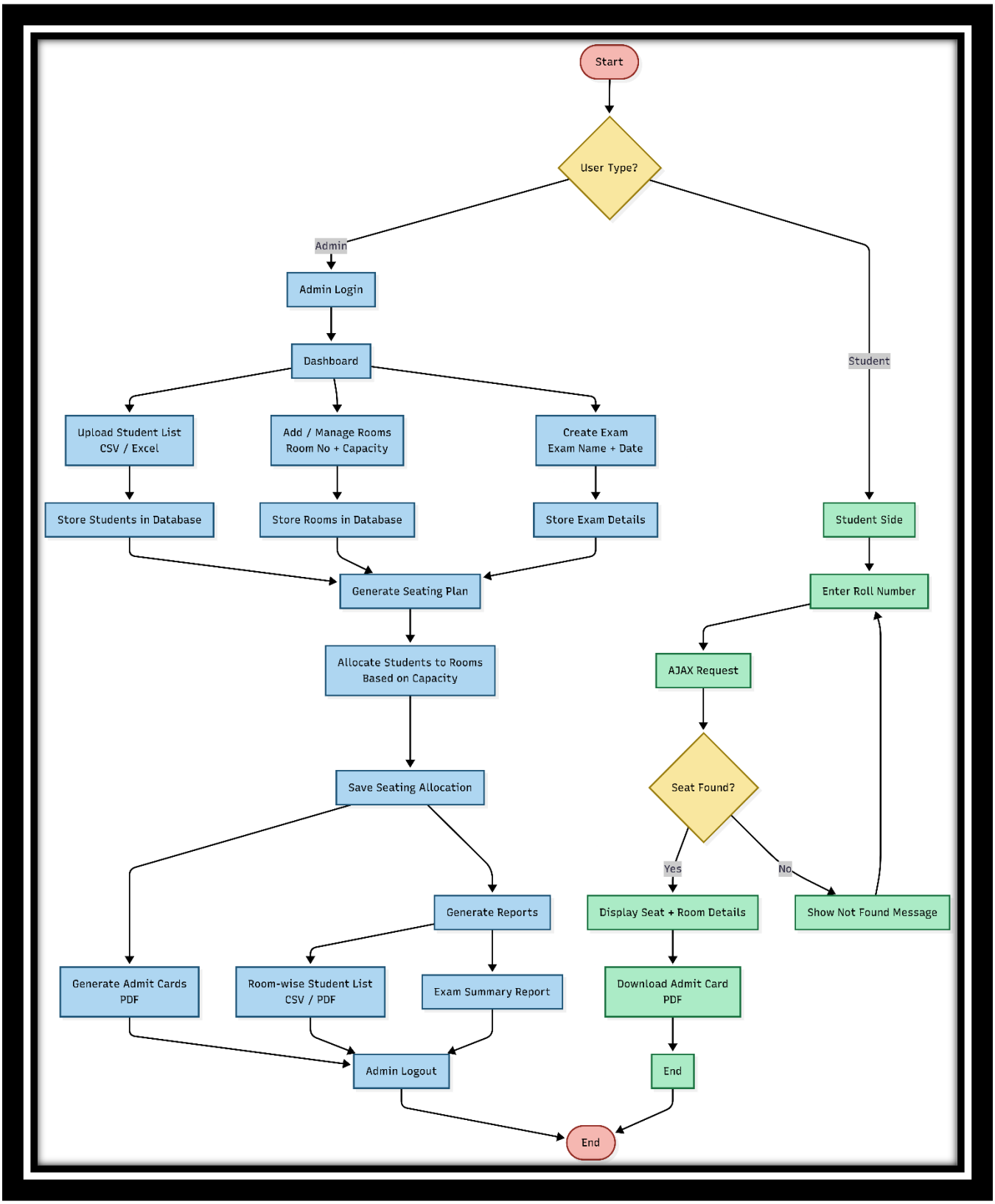
2. Project Features

- Admin can upload student list (CSV and Excel)
- Admin can add and manage exam rooms with capacity
- Automatic seating plan generation
- Individual Admit Card generation in PDF format
- Student seat search using AJAX (without page reload)
- Room-wise student list reports (CSV and PDF)
- Secure admin authentication using Django

3. Proposed Methodology

- Frontend: HTML, CSS, Bootstrap, JavaScript (AJAX)
- Backend: Python and Django (MVT architecture)
- Database: SQLite using Django ORM
- Models: Student, Room, Exam, SeatingAllocation (with relationships)
- Forms and ModelForms for data input and validation
- Django Authentication for secure admin login
- PDF generation using ReportLab and CSV export using Django tools

4. Work Flow Diagram



5. Relevant Full Stack Concepts Used

- MVT Architecture : Model-View-Template separation in Django
- Django ORM : Creating models and handling database operations
- Model Relationships : One-to-Many and Many-to-Many relations (Student Room Exam)
- CRUD Operations : Create, Read, Update, Delete using Django views
- Forms and ModelForms : Data input and validation
- URL Routing : Mapping URLs to views
- Template Inheritance : Reusable base templates and static files
- Authentication : Secure admin login using Django Auth
- AJAX : Asynchronous seat search without page reload
- Report Generation : Exporting data to CSV and PDF

6. Expected Outcomes

- A fully functional mini full-stack web application.
- Automatic and accurate seating arrangement generation.
- Downloadable admit cards in PDF format for each student.
- Fast real-time seat search without page reload using AJAX.
- Useful room-wise and exam-related reports for the examination cell.

7. Technology Stack

- Language: Python
- Framework: Django
- Frontend: HTML, CSS, Bootstrap, JavaScript
- Database: SQLite
- Additional Libraries: ReportLab for PDF , Django CSV export

8. Conclusion

The Exam Management System successfully integrates multiple full-stack development concepts to provide an efficient and automated solution for examination management. By combining techniques such as Django ORM for database handling, ModelForms for data validation, and AJAX for real-time seat searching, the system is able to generate accurate seating arrangements and individual admit cards effectively.

The use of template inheritance and static file management improves the structure and reusability of the web interface, while CSV and PDF report generation enhances the practical utility of the system for examination cells. The Django authentication system ensures secure access for administrators.

Overall, the system demonstrates a practical application of full-stack development concepts in solving real-world academic problems. It is modular, scalable, and can be further enhanced with advanced features such as email notifications, QR code verification, and multi-exam support.